**QPCODE:** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM
# FOURTH SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION

## Fourth Semester
## MG4DSECFS202-DISTRIBUTED SYSTEMS

(2024 ADMISSION ONWARDS)

**Duration: 2 Hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

## Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Define Distributed System. [K] / [1]

2. Explain the term resource sharing in distributed systems. [U] / [1]

3. Explain multiprocessor systems. [U] / [1]

4. Differentiate between homogeneous and heterogeneous distributed systems. [An] / [3]

5. Explain Distributed Operating System. [U] / [1]

6. Illustrate Remote Procedure Call (RPC) with a simple example. [An] / [3]

| | | |
|---|---|---|
| 7. | Define MOM. | [K] / [1] |
| 8. | Explain message-oriented communication. | [U] / [1] |
| 9. | Define multithreading in distributed systems. | [K] / [1] |
| 10. | Illustrate code migration in distributed systems. | [An] / [3] |
| 11. | Define logical clocks. | [K] / [1] |
| 12. | Define fault tolerance in distributed systems. | [K] / [1] |

**[2x10 = 20]**

## Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

| | | |
|---|---|---|
| 13. | Explain the characteristics of distributed systems with suitable examples. | [U] / [1] |
| 14. | Compare multiprocessor systems and multicomputer systems. | [An] / [3] |
| 15. | Differentiate between uniprocessor operating systems and distributed operating systems. | [An] / [3] |
| 16. | Explain the working of RPC and give its advantages. | [U] / [1] |
| 17. | Compare Remote Method Invocation (RMI) and RPC. | [An] / [3] |
| 18. | Evaluate message-oriented middleware and explain its role in distributed systems. | [E] / [4] |
| 19. | Differentiate between persistent communication and transient communication. | [An] / [3] |
| 20. | Evaluate the concept of multithreaded servers with the help of a neat diagram. | [E] / [4] |

**[5x6 = 30]**

## Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

| | | |
|---|---|---|
| 21. | Illustrate the different communication models in distributed systems including RPC, RMI, and message - oriented communication. | [An] / [3] |
| 22. | Evaluate and compare centralized mutual exclusion, distributed mutual exclusion, and token-ring algorithms with suitable examples. | [E] / [4] |
| 23. | Evaluate different replication techniques and their role in fault tolerance. | [E] / [4] |

**[10x2 = 20]**